

Custody

Proving digital evidence has not been changed

ICSC 2026 Universities Hackathon, Track H
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The problem

Cases collapse because nobody can prove the evidence was not changed.

Not because anyone changed it.

Evidence goes onto a flash drive, gets emailed between officers, sits on a shared machine, and arrives at a hearing a year later. Maybe nobody touched it. Nobody can prove that either.

Where this happens

Three settings

Cybercrime cases

phone extractions, server logs

Fraud investigations

transaction records, audit logs

Disciplinary panels

message exports, access logs

In each one, the handling record is on paper, if it exists at all.

The idea

A computer can read a file and produce a short code from its contents.

Same file, same code, every time.

Change one letter anywhere inside, and the code comes out completely different.

We take that code at the scene, on the collecting officer's own phone, before the file goes anywhere else. We take it again months later and compare. Match means untouched.

**Every entry is knotted to the one before it,
like beads on a string.**

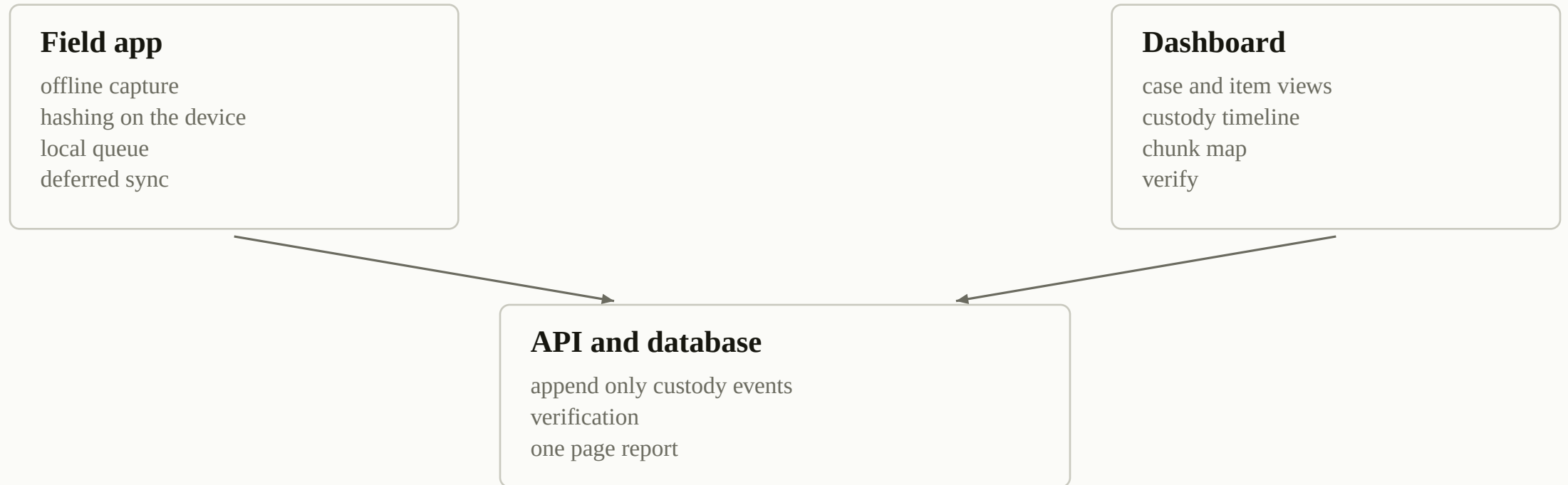


pull one out, and the string breaks here

You cannot quietly remove a bead from the middle.

How it fits together

Architecture

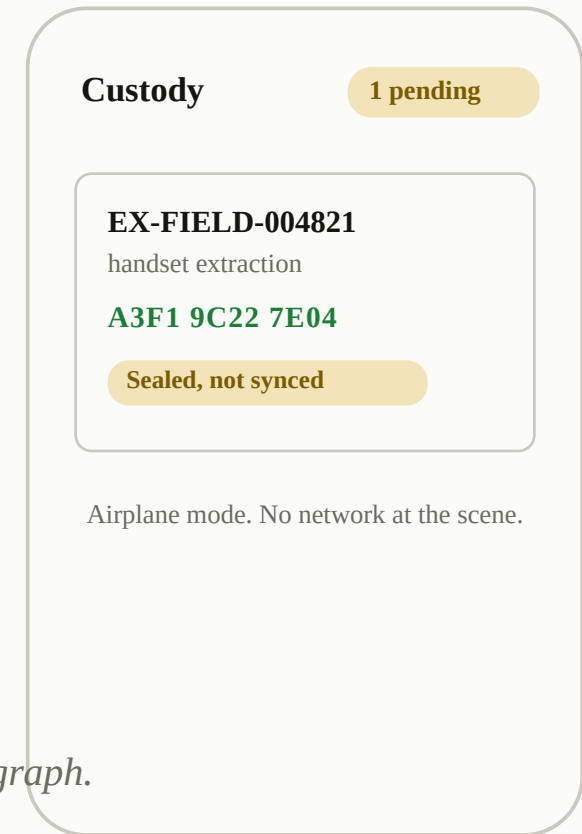


Everything runs on one laptop. No cloud. Nothing paid for.

**The fingerprint is taken at the scene,
on the officer's device, before the file
goes anywhere.**

Nothing between the scene and the server can alter the file
without it being detectable.

This is a question about power cuts and dead networks answered by architecture, not by a paragraph.



One extraction can exceed 100GB

We say which part changed, not just that something did.



The file is hashed in 4MB pieces. Part 6 of 8 differs, so the change lies between byte 20,971,520 and byte 25,165,823.

We tested at 30MB and extrapolated. Memory use stays flat as the file grows, because we never hold the whole file at once. That is what makes the claim reasonable, not the size of our test file.

Demo

The deliverable

One page. No jargon. No hashes.

Is it unchanged? Yes.

How do we know? A fingerprint was taken at collection and checked again.

Who has handled it? Four people, named, with dates.

Can this list have been edited? No, and here is why.

"A process that is technically perfect but cannot be explained in a hearing has not solved the problem."

the brief

[Replace this slide with a photograph of the printed report in someone's hands.]

We tested it on people

We gave the report to someone outside computer science.

[What confused them.]

[What we changed as a result.]

Fill this in with the real result before presenting. Do not invent it. Almost no other team will have this slide, and it is worth more than anything we could make up.

What we did not solve

Honest limits

A corrupt collector

we protect the chain from collection onward, not before it

The device clock

we show both clocks, we do not vouch for the phone's

Scale

tested at 30MB, extrapolated, not a real 100GB extraction

Per officer signing

events are tied to a login, not to a person's key

Storage

we prove change, we do not prevent deletion

Bulk compromise

full server access could rebuild the whole chain

If asked

Why not a blockchain?

A blockchain solves distrust between parties with no shared authority.

A court has one.

A hash chain, append only storage, and periodically published root hashes give the same tamper evidence at a fraction of the complexity, and can be explained to a panel member in a single sentence.